Supplementary Material

Physics-Guided Probabilistic Glucose Forecasting in Type 1 Diabetes: An Integrity-Checked Multi-Horizon Evaluation with a Stable Patient-Specific Insulin-Sensitivity Estimate

Notation, Variables, and Terminology

Rate constants are expressed per minute unless stated otherwise. A subscript b denotes a basal or fasting quantity, a hat denotes a prediction, bold lowercase symbols denote vectors, and bold uppercase symbols denote matrices. Bracketed reference numbers refer to the reference list of the main text. The patient-parameter vector θ_p is listed in Table XV of the main text.

Table S1: Physiological state variables and external inputs

Symbol	Unit	Name	Meaning in this study
$G(t)$	mg/dL	Plasma glucose	Glucose concentration forecast by the learned spline and constrained by the mechanistic equation.
G_b	mg/dL	Basal glucose	Subject-specific fasting anchor used by the Bergman equilibrium.
G_0	mg/dL	Anchor glucose	Last real CGM observation at forecast time $t = 0$.
$X(t)$	min^{-1}	Remote insulin action	Interstitial insulin effect entering glucose disposal additively with p_1.
$I(t)$	μU/mL	Plasma insulin	Circulating insulin concentration generated by subcutaneous absorption and clearance.
I_b	μU/mL	Basal plasma insulin	Concentration implied by the subject's basal delivery; subtracting it preserves basal equilibrium.
S_1, S_2	U	SC insulin states	First and second subcutaneous absorption compartments.
Q_{sto}	g	Stomach carbohydrate	Ingested carbohydrate awaiting gastric emptying.
Q_{gut}	g	Gut carbohydrate	Carbohydrate available for intestinal absorption.
$R_a(t)$	mg/min	Glucose appearance rate	Meal-derived glucose entering the accessible glucose compartment.
u_{ins}	U/min	Insulin input	Basal plus bolus insulin delivery on the five-minute grid.
u_{carb}	g/min	Meal input	Recorded carbohydrate ingestion on the grid.
$\mathbf{z}$	–	Compartment state	$[S_1, S_2, I, Q_{\text{sto}}, Q_{\text{gut}}, X]^\mathsf{T}$.
$\mathbf{u}$	–	Input vector	$[u_{\text{ins}}, u_{\text{carb}}, I_b]^\mathsf{T}$.

Table S2: Architecture, discretization, and uncertainty notation

Symbol	Value or dimension	Definition
t	0–120 min	Time measured from the forecast anchor.
h	30, 60, 90, 120 min	Forecast horizon.
Δ	5 min	Common temporal grid step.
B	batch dependent	Number of windows processed together.
H	4	Number of reported forecast horizons.
K	12	Number of cubic B-spline basis functions.
$B_k(t)$	spline basis	The kth clamped, uniformly knotted B-spline basis function.
c_k	learned scalar	Spline coefficient emitted by the trajectory head.
$G_\theta(t)$	mg/dL	Anchored learned trajectory $G_0 + \sum_k c_k[B_k(t) - B_k(0)]$.
A, B	6×6, 6×3	Continuous compartment-state and input matrices.
A_d, B_d	discrete matrices	Exact zero-order-hold matrices obtained from one augmented matrix exponential.
θ_p	10 elements	Patient-specific physiological parameter vector in Table XV of the main text.
$g = \sigma(\gamma)$	initialized near 0.10	Learned trust gate controlling the mechanistic prior.
$\hat{G}_{0.5}$	mg/dL	Median forecast and reported point prediction.
$\hat{G}_q$	mg/dL	Forecast at quantile level $q \in \{0.1, 0.5, 0.9\}$.
$\delta_q(h)$	mg/dL	Positive softplus-derived distance from the median, preventing quantile crossing.
$r(t)$	dimensionless	Scale-normalized physics residual evaluated at 121 collocation points.
n_{eff}	$\approx N/48$	Approximate independent sample count after accounting for window overlap.

Table S3: Training-loss terminology

Term	Meaning
$\mathcal{L}_{\text{Huber}}$	Horizon-weighted robust point loss for the median forecast; quadratic near zero error and linear for large errors.
$\mathcal{L}_{\text{pinball}}$	Proper quantile loss $\max\{q(y-\hat{y}_q),(q-1)(y-\hat{y}_q)\}$.
$\mathcal{L}_{\text{phys}}$	Mean squared, nondimensionalized Bergman residual over the collocation grid.
$\mathcal{L}_{\text{prior}}$	Weak population-prior penalty that discourages physiologically implausible parameter drift.
$\mathcal{L}_{\text{temporal}}$	Within-subject penalty on disagreement between parameter estimates from adjacent windows.
λ_{phys}	Physics weight, adaptive in the relevant ablation arms and optionally multiplied by a curriculum ramp.
$s_i = \log \sigma_i^2$	Stable log-variance parameterization used for learned multi-objective weighting [51].

Table S4: Evaluation metrics and clinical terms

Term	Unit	Definition and interpretation
MAE	mg/dL	Mean absolute error, $n^{-1}\sum_i \lvert\hat{y}_i - y_i\rvert$.
RMSE	mg/dL	Root-mean-square error, emphasizing larger deviations.
R^2	–	$1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$.
MARD	%	Mean absolute relative difference, with reference glucose in the denominator.
Skill	%	$1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{persistence}}$. Positive values beat no-change prediction.
TIR/TAR/TBR	%	Time in, above, or below the consensus glucose range. The main in-range interval is 70–180 mg/dL [55].
CV ratio	–	$\text{CV}_{\text{pred}}/\text{CV}_{\text{actual}}$; values below one quantify excursion compression.
LBGI/HBGI	–	Low/high blood-glucose risk indices after the Kovatchev symmetric risk transform.
Coverage	–	Fraction of observations below a stated predictive quantile; calibration requires observed and nominal coverage to agree.
Hypo sensitivity	–	Fraction of actual $G < 70$ mg/dL events detected by the forecast or alarm.
Precision	–	Fraction of hypoglycemia alarms corresponding to actual events.
ICC(1)	–	One-way, single-measure intraclass correlation used for S_I test–retest stability [60].
ρ	–	Spearman rank correlation used for monotonic association with insulin dose [61].
IG_i	mg/dL	Integrated-gradient attribution of feature i relative to the standardized-mean baseline.
Clarke/Parkes EGA	zone	Point-wise clinical error grids originally designed for glucose-measurement error.
PRED-EGA	category	Prediction-specific error grid combining level and rate information [69]; reported in the Results of the main text.

Table S5: Abbreviations and protocol terminology

Short form	Expansion	Study-specific meaning
T1D/IDDM	Type 1 diabetes / insulin-dependent diabetes mellitus	IDDM is retained only when describing older physiological literature.
CGM	Continuous glucose monitoring	Nominal five-minute glucose sampling in OhioT1DM.
SC	Subcutaneous	Route represented by insulin compartments S_1 and S_2.
IOB/COB	Insulin/carbohydrate on board	Derived as $S_1 + S_2$ and $Q_{\text{sto}} + Q_{\text{gut}}$.
TDD/ICR	Total daily insulin dose / insulin-to-carbohydrate ratio	External consistency variables for the physiology analysis; ICR is available only for six subjects.
PINN/ODE	Physics-informed neural network / ordinary differential equation	Neural training constrained by the physiological differential equations.
LOSO	Leave-one-subject-out	Twelve subject-disjoint folds, each withholding all data from one test subject.
Official protocol	OhioT1DM temporal holdout	Later data from subjects whose earlier data are available for training; therefore personalized.
BGLP	Blood Glucose Level Prediction challenge	OhioT1DM challenge setting; 2018 and 2020 have different test warmup rules.
EGA	Error-grid analysis	Clarke and Parkes clinical zone classification.
IG/SHAP	Integrated gradients / SHapley Additive exPlanations	IG is used; the legacy single-timestep SHAP procedure was rejected.
SD/CI/ICC	Standard deviation / confidence interval / intraclass correlation	Uncertainty and reliability summaries computed at the subject level.
A0–A7	Ablation arms	Controlled variants of physics form, weighting, parameter personalization, and curriculum.
ZOH	Zero-order hold	Exact interval propagation under piecewise-constant inputs; also used in the literature for persistence.

Secondary Analyses of the Physics Contribution

These analyses compare A7 (physics-guided) with A0 (no physics) on the official test windows, with the five fair-selection seeds of Table VIII in the main text. They were written down and time-stamped before they were run (`PREREG_ROUND6.md`), after the main results were known; deviations are listed in `ROUND6_RESULTS.md`. Gap = A0 − A7 (positive = A7 better), seed-averaged per subject; paired Wilcoxon test and subject-bootstrap 95% CI (5,000 resamples). n.d. = no difference detected.

Table S6: Secondary analyses of the A0–A7 MAE gap (mg/dL; positive = A7 better), seed-averaged per subject over five seeds. Holm correction over all 55 tests in this family; with 12 subjects the smallest attainable Holm-adjusted p is about 0.027. P1: horizon trend; P2: post-meal (0–180 min after a carbohydrate entry) vs. other windows; P3: target glycemic range; P4: rate of change at the forecast origin (mg/dL/min); P5: fraction of forecast trajectories with $|dG/dt| > 4$ mg/dL/min or outside 40–400 mg/dL; P7: across-seed SD of mean MAE.

	Stratum	h	n	A0	A7	Gap	95% CI	raw p	Holm p	Verdict
P1	all	30	12	13.54	13.37	+0.17	[+0.06, +0.30]	0.0093	0.353	n.d.
P1	all	60	12	23.01	22.60	+0.41	[+0.18, +0.67]	0.0049	0.205	n.d.
P1	all	90	12	30.01	29.28	+0.72	[+0.37, +1.11]	0.0024	0.110	n.d.
P1	all	120	12	35.07	34.08	+0.99	[+0.53, +1.49]	0.0015	0.069	n.d.
P1	trend (per-subject Spearman gap horizon, Wilcoxon vs 0)	–	12	–	–	+0.68	[+0.33, +0.97]	0.0039	0.168	n.d.
P2	post-meal (0-180 min)	30	11	16.38	16.22	+0.16	[-0.08, +0.44]	0.4648	1.000	n.d.
P2	other	30	12	11.74	11.59	+0.15	[+0.08, +0.23]	0.0010	0.049	helps
P2	difference of gaps (post-meal minus other)	30	11	–	–	+0.01	[-0.21, +0.26]	0.8984	1.000	n.d.
P2	post-meal (0-180 min)	60	11	26.79	26.60	+0.19	[-0.43, +0.80]	0.6377	1.000	n.d.
P2	other	60	12	20.49	20.02	+0.47	[+0.32, +0.65]	0.0005	0.027	helps
P2	difference of gaps (post-meal minus other)	60	11	–	–	-0.30	[-0.89, +0.26]	0.5771	1.000	n.d.
P2	post-meal (0-180 min)	90	11	33.77	33.47	+0.30	[-0.33, +0.89]	0.4648	1.000	n.d.
P2	other	90	12	27.56	26.59	+0.97	[+0.61, +1.36]	0.0005	0.027	helps
P2	difference of gaps (post-meal minus other)	90	11	–	–	-0.73	[-1.28, -0.23]	0.0244	0.781	n.d.
P2	post-meal (0-180 min)	120	11	38.07	37.53	+0.54	[-0.04, +1.10]	0.1230	1.000	n.d.
P2	other	120	12	33.05	31.69	+1.36	[+0.72, +2.07]	0.0024	0.110	n.d.
P2	difference of gaps (post-meal minus other)	120	11	–	–	-0.94	[-1.81, -0.11]	0.0537	1.000	n.d.
P3	target <70	30	12	13.44	13.19	+0.25	[-0.87, +1.34]	0.5693	1.000	n.d.
P3	target 70-180	30	12	11.98	11.89	+0.09	[-0.03, +0.21]	0.2334	1.000	n.d.
P3	target >180	30	12	16.97	16.67	+0.30	[+0.12, +0.47]	0.0122	0.439	n.d.
P3	target <70	60	12	30.17	29.38	+0.79	[-1.77, +3.11]	0.3804	1.000	n.d.
P3	target 70-180	60	12	19.58	19.51	+0.07	[-0.25, +0.40]	0.9697	1.000	n.d.
P3	target >180	60	12	29.85	29.00	+0.85	[+0.51, +1.24]	0.0005	0.027	helps
P3	target <70	90	12	43.18	43.11	+0.07	[-3.56, +3.08]	0.6221	1.000	n.d.
P3	target 70-180	90	12	24.71	24.61	+0.10	[-0.39, +0.54]	0.5693	1.000	n.d.
P3	target >180	90	12	40.18	38.60	+1.58	[+0.89, +2.32]	0.0010	0.049	helps
P3	target <70	120	12	52.57	52.69	-0.12	[-4.34, +3.37]	0.6221	1.000	n.d.
P3	target 70-180	120	12	28.28	27.95	+0.33	[-0.23, +0.82]	0.1763	1.000	n.d.
P3	target >180	120	12	47.73	46.00	+1.73	[+0.59, +2.99]	0.0210	0.693	n.d.
P4	|ROC_15| <1	30	12	11.86	11.71	+0.15	[+0.08, +0.23]	0.0068	0.273	n.d.
P4	|ROC_15| 1-2	30	12	18.26	18.05	+0.21	[-0.12, +0.59]	0.4697	1.000	n.d.
P4	|ROC_15| >2	30	12	25.08	24.87	+0.21	[-0.33, +0.74]	0.5693	1.000	n.d.
P4	|ROC_15| <1	60	12	20.96	20.53	+0.43	[+0.24, +0.63]	0.0015	0.069	n.d.
P4	|ROC_15| 1-2	60	12	29.23	28.89	+0.33	[-0.28, +1.03]	0.4697	1.000	n.d.
P4	|ROC_15| >2	60	12	35.84	35.71	+0.13	[-0.84, +1.06]	0.7334	1.000	n.d.
P4	|ROC_15| <1	90	12	27.98	27.21	+0.78	[+0.43, +1.14]	0.0005	0.027	helps
P4	|ROC_15| 1-2	90	12	36.13	35.53	+0.60	[-0.09, +1.25]	0.2036	1.000	n.d.
P4	|ROC_15| >2	90	12	43.02	42.73	+0.29	[-1.20, +1.78]	0.7910	1.000	n.d.
P4	|ROC_15| <1	120	12	33.09	31.98	+1.12	[+0.65, +1.61]	0.0010	0.049	helps
P4	|ROC_15| 1-2	120	12	41.00	40.42	+0.59	[-0.16, +1.35]	0.2036	1.000	n.d.
P4	|ROC_15| >2	120	12	47.30	47.06	+0.24	[-1.25, +1.72]	0.9697	1.000	n.d.
P5	frac trajectories with —dG/dt—>4 mg/dL/min	–	12	0.368	0.000	+0.368	[+0.345, +0.390]	0.0005	0.027	helps
P5	frac trajectories <40 or >400 mg/dL	–	12	0.000	0.000	+0.000	[-0.000, +0.001]	0.6250	1.000	n.d.
P7	across-seed SD of mean MAE (gap = SD_A0 - SD_A7)	30	5	0.097	0.114	-0.017	[-0.114, +0.072]	0.4262	1.000	n.d.
P7	across-seed SD of mean MAE (gap = SD_A0 - SD_A7)	60	5	0.160	0.320	-0.160	[-0.289, +0.060]	0.1038	1.000	n.d.
P7	across-seed SD of mean MAE (gap = SD_A0 - SD_A7)	90	5	0.152	0.527	-0.376	[-0.571, -0.045]	0.0326	0.978	n.d.
P7	across-seed SD of mean MAE (gap = SD_A0 - SD_A7)	120	5	0.101	0.637	-0.536	[-0.673, -0.097]	0.0187	0.637	n.d.

In a post hoc check of P5 (not part of the family above), the A0 excursions lay almost entirely in the first and last 10 min of the 2-h trajectory: excluding those minutes, 0.10% of A0 trajectories and 0% of A7 trajectories exceeded 4 mg/dL/min.

Table S7: Reduced training data (P6): A0 and A7 retrained on 25% or 50% of the training subjects (three seeds each; same test set). MAE in mg/dL; gap = A0 − A7. The 25% and 50% rows are in the Holm family of Table S6; the 100% rows (same three seeds) are a descriptive reference only.

Training data	h	A0	A7	Gap	95% CI	raw p	Holm p
25%	30	16.18	15.69	+0.49	[-0.09, +0.92]	0.0425	1.000
25%	60	26.79	25.54	+1.25	[+0.38, +1.99]	0.0269	0.833
25%	90	34.29	32.35	+1.94	[+0.88, +2.93]	0.0122	0.439
25%	120	39.72	37.37	+2.35	[+1.21, +3.46]	0.0049	0.205
50%	30	14.87	14.72	+0.15	[-0.01, +0.32]	0.2661	1.000
50%	60	24.95	24.40	+0.55	[+0.16, +0.97]	0.0342	0.991
50%	90	32.43	31.45	+0.99	[+0.41, +1.53]	0.0093	0.353
50%	120	37.96	36.55	+1.41	[+0.64, +2.13]	0.0068	0.273
100% (descriptive)	30	13.50	13.36	+0.14	[-0.01, +0.29]	0.0522	–
100% (descriptive)	60	22.93	22.61	+0.32	[+0.05, +0.62]	0.0640	–
100% (descriptive)	90	29.95	29.34	+0.61	[+0.21, +1.03]	0.0161	–
100% (descriptive)	120	35.07	34.12	+0.94	[+0.48, +1.41]	0.0034	–